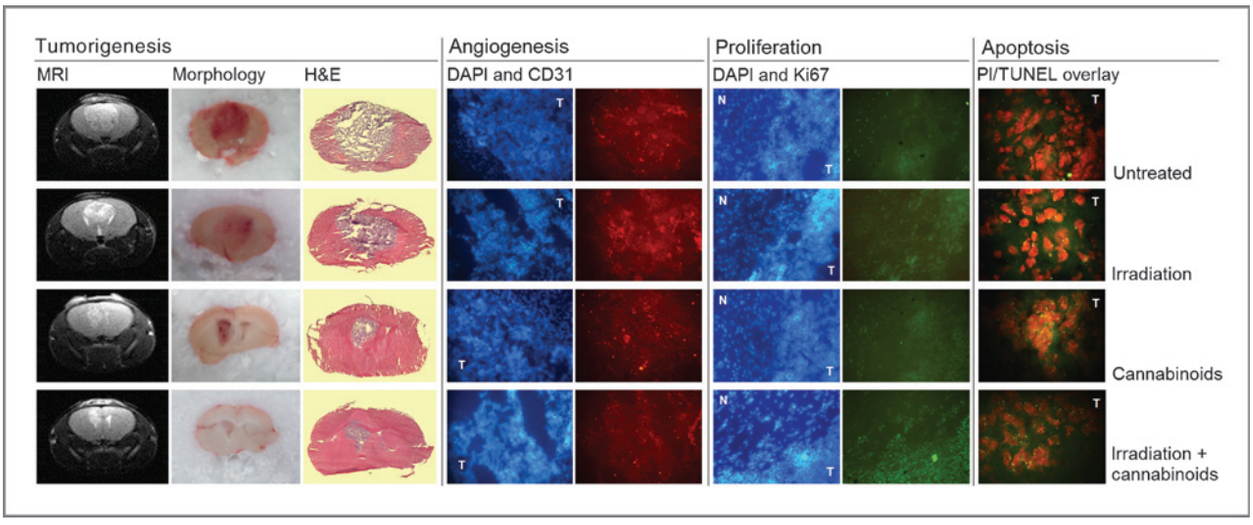

Figure 5. Effect of CBD+THC and irradiation on tumor morphology. Sections of the brains removed from mice treated with CBD+THC, irradiation and CBD+THC with irradiation had tumor masses that were distinguishable *in situ*. Images, where T denotes tumor region and N denotes normal tissue, were: MRI images obtained on day 21; morphology of whole-brain tissue, images of a coronal slice of the brain in mountant; composite H&E; angiogenesis: DAPI (blue) and CD31 (red); proliferation: DAPI (blue) and Ki67 (green); apoptosis: overlay of propidium iodide (PI; red) and TUNEL (green).

showed that 4 Gy irradiation had no dramatic effect on tumor growth, whereas CBD and THC administered together reduced tumor progression (Fig. 4A and B). Combining the cannabinoids with irradiation impeded even further the rate at which tumor growth progressed, which was virtually stagnant throughout the experiment course (Fig. 4B); and correspondingly, tumor sizes on the final day of the study were significantly smaller in this cohort of animals compared with any of the others (Fig. 4C). The ensuing *ex vivo* analyses of the brains resected from the mice showed that *in situ* the tumor masses were denser in the two groups that had been treated with the cannabinoids, and furthermore, that these tumors also appeared less sanguine. Subsequent immunohistochemical analysis confirmed this: expression of the vascularization marker CD31 was lower in the cannabinoid treated groups, particularly in the combination treated samples. Expression of the proliferation marker Ki67 appeared similar in all samples (Fig. 5). TUNEL staining was positive in all sections, but generally higher where cannabinoids were used, and specifically when combined with irradiation. Mean staining scores for TUNEL were 0/+, +, +, and +/++ for controls, IR, cannabinoids, and the combination groups, respectively (Fig. 5).

Discussion

There is ever-increasing evidence supporting a role for cannabinoids in cancer therapy, and so the current study was undertaken to explore this further. Of particular interest is the impact that cannabinoids have on signaling pathways that underlie cell fate. This impact allows scope for exciting combination strategies, specifically, the effect of combining these agents with existing therapeutic modalities, which presents an attractive novel avenue for research. In this study, we have concentrated on combining cannabinoids with radiotherapy as this is the primary form of treatment for patients with glioma following surgical resection. The principal aim of the current study, therefore, was to assess the efficacy of CBD and THC in combination with radiotherapy, to determine the potential clinical benefits of this combination in improving radiotherapy outcome. The most significant finding of this study has been the dramatic reduction in tumor growth *in vivo* when cannabinoids and ionizing radiation were used together.

CBD and THC were tested in this study as they have previously been shown to be effective in glioma both *in vitro* and *in vivo*, and display multiple mechanisms of action. Both cannabinoids can reduce cell numbers by inhibiting cell-cycle progression and cell growth as well as by triggering apoptosis and engaging autophagy (19), and are also antiangiogenic and antimigratory (15). The two compounds have also been combined in a preparation that is currently licensed to treat multiple sclerosis, which is now undergoing trials in patients with glioma. In the first part of the study, we assessed the individual activities of CBD and THC in two human glioma cell lines, and confirmed that they both cause dose-dependent reductions in cell numbers in these lines. These reductions were associated with mild increases in apoptosis and no clear alterations to the DNA profiles of the cells. This suggested the cannabinoids were both cytotoxic and cytostatic in nature under our *in vitro* test conditions.

In the preparation of the compounds used in the current study, the cannabinoids were extracted and purified from the cannabis plant. Depending on the level of purification, this process can result in CBD and THC preparations that contain small amounts of other

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